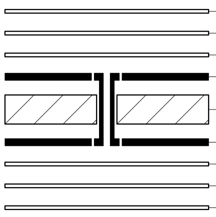
3ART13 LED Fabrication Document

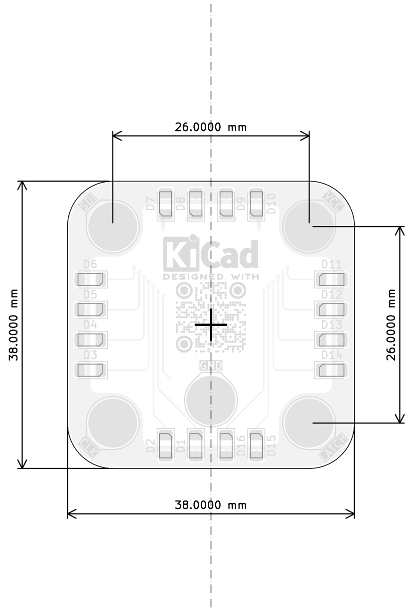
Layer Stack Legend



Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
Copper	**T.Cu (Sig, PWR)**	**0.035mm (1.00oz)**		**Signal**	**GBR**
Core		1.5mm	FR4_7628	Dielectric	
Copper	**B.Cu (Sig, PWR)**	**0.035mm (1.00oz)**		**Signal**	**GBR**
	B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1.61mm
Note: external layer thicknesses are specified after plating

Top Fabrication (Scale 1:1)



Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0.2032	0.28	L2

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

4) SURFACE FINISH: IMMERSION GOLD

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
B. Tg 170 C OR EQUIVALENT.
C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY BAJRAN IMGAGINARY INDUSTRIES.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	38.000 × 38.000 mm
BOARD THICKNESS	1.610 mm
TRACE WIDTH	0.200 mm
TRACE TO TRACE	0.200 mm
MIN. HOLE (PTH)	0.250 mm
MIN. HOLE (NPTH)	N/A mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.254 mm
COPPER TO EDGE	0.250 mm
HOLE TO HOLE	0.254 mm

11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.

12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

Comments:

Company: **BaJRan imgaginary industries**

Variant: RELEASED

Git Hash: 965b332

Board Name: **3ART13 LED**

Project Name: **Sienkiewicz**

Sheet Title: Top Fabrication (Scale 1:1)

File Name: Sienkiewicz_LED.kicad_pcb

Designer: Pawel Baran

Date: 2024-04-13

Revision: 2.1.1

Sheet Path:

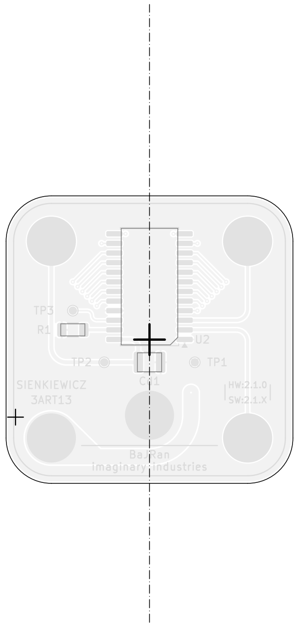
Reviewer:

Size: **A4**

Sheet: **1** of **7**

3ART13 LED Fabrication Document

Bottom Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

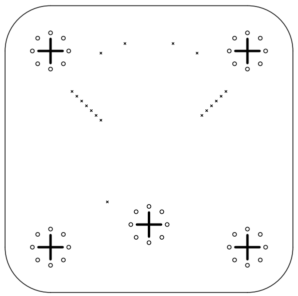
Comments:	Company:		Variant:	Git Hash:
	BaJRan imgaginary industries		RELEASED	965b332
	Board Name: **3ART13 LED**		Project Name: **Sienkiewicz**	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Sienkiewicz_LED.kicad_pcb	Designer: Pawel Baran	Date: 2024-04-13	Revision: 2.1.1
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **2** of **7**

3ART13 LED Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	18	0.25mm (9.84mils)	PTH	Round	T.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Via
○	40	0.50mm (19.69mils)	PTH	Round	T.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
+	5	3.20mm (125.98mils)	PTH	Round	T.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
	Total 63					

Drill Drawing L1 - L2 (Scale 1:1)



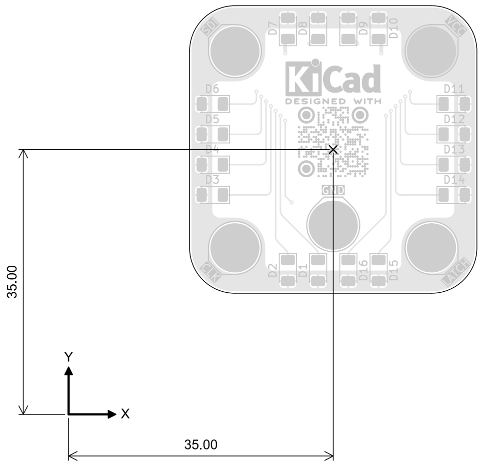
Comments:	Company: BaJRan imgaginary industries		Variant: RELEASED	Git Hash: 965b332
	Board Name: 3ART13 LED		Project Name: Sienkiewicz	
Sheet Title: Drill Drawing (L1 - L2)	File Name: Sienkiewicz_LED.kicad_pcb	Designer: Pawel Baran	Date: 2024-04-13	Revision: 2.1.1
Sheet Path:		Reviewer:	Size: A4	Sheet: 3 of 7

3ART13 LED Fabrication Document

Top Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]

Ref.	Net	X [mm]	Y [mm]



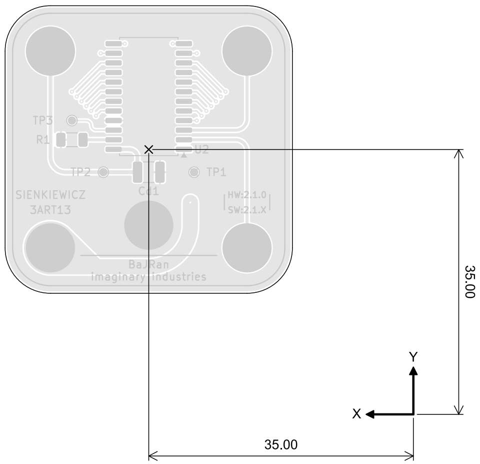
All dimensions are in millimeters unless otherwise specified.

Comments:	Company: BaJRan imgaginary industries		Variant: RELEASED	Git Hash: 965b332
	Board Name: **3ART13 LED**		Project Name: **Sienkiewicz**	
Sheet Title: Top Test Points (Scale 1:1)	File Name: Sienkiewicz_LED.kicad_pcb	Designer: Pawel Baran	Date: 2024-04-13	Revision: 2.1.1
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **4** of **7**

3ART13 LED Fabrication Document

Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
TP1	GND	78.50	13.00
TP2	+5V	90.50	13.00
TP3	Net-(U2-SDO)	94.66	19.82



All dimensions are in millimeters unless otherwise specified.

Comments:	Company: BaJRan imgaginary industries		Variant: RELEASED	Git Hash: 965b332
	Board Name: 3ART13 LED		Project Name: Sienkiewicz	
Sheet Title: Bottom Test Points (Scale 1:1)	File Name: Sienkiewicz_LED.kicad_pcb	Designer: Pawel Baran	Date: 2024-04-13	Revision: 2.1.1
Sheet Path:		Reviewer:	Size: A4	Sheet: 5 of 7

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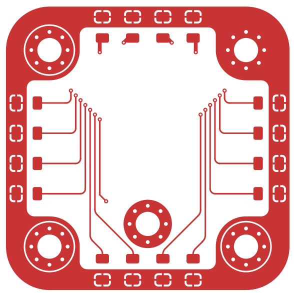
T.Cu (Sig, PWR) (Scale 1:1)



Comments:	Company: BaJRan imgaginary industries		Variant: RELEASED	Git Hash: 965b332
	Board Name: **3ART13 LED**		Project Name: **Sienkiewicz**	
Sheet Title: T.Cu (Sig, PWR) (Scale 1:1)	File Name: Sienkiewicz_LED.kicad_pcb	Designer: Pawel Baran	Date: 2024-04-13	Revision: 2.1.1
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **6** of **7**

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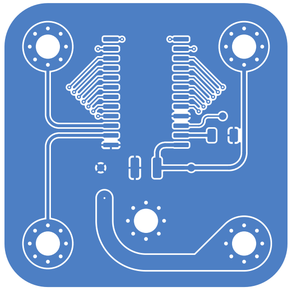
B.Cu (Sig, PWR) (Scale 1:1)



Comments:	Company: BaJRan imgaginary industries	Variant: RELEASED	Git Hash: 965b332
	Board Name: 3ART13 LED	Project Name: Sienkiewicz	
Sheet Title: B.Cu (Sig, PWR) (Scale 1:1)	File Name: Sienkiewicz_LED.kicad_pcb	Designer: Pawel Baran	Date: 2024-04-13	Revision: 2.1.1
Sheet Path:		Reviewer:	Size: A4	Sheet: 7 of 7